Chapter (23):- Electric Fields:-

23.1 Properties of charges:-

1- Kinds of charges :- +ve and -ve charges

Charges of the same sign repel each other and opposite sign attract each other.

2- Conservation of electric charges:-

when one object rubbed against another, one object gains +ve charge while the other gains an equal amount of -ve charge like rod of glass (+ve) rubbed by silk (-ve).

the total charge on isolated system is conserved.

3- Quantization of electric charge :-

charge appears as some integral multiple of a fundamental amount of charge (e)

$q = ne$ [n :- integer, e :- charge of electron $e = 1.6 \times 10^{-19}$ C]

23.2 Charging objects by induction :-

materials classified according to ability of electrons to move through it to :-

a- Conductors :- materials in which some of the electrons are free electrons that are not bound to atom and can move freely through the material.

b- insulators :- materials in which all electrons are bound to atoms and cannot move freely through the material.

c- semi conductors :- their electrical properties are between insulators and conductors.

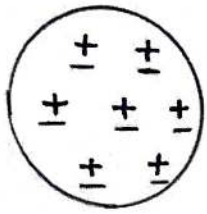
methods of charging :-

1- charging by contact.

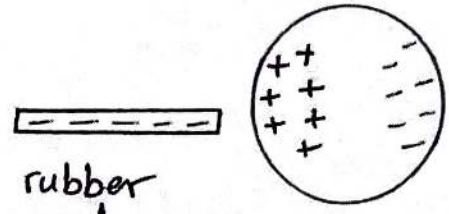
2- charging by rubbing.

3- charging by induction.

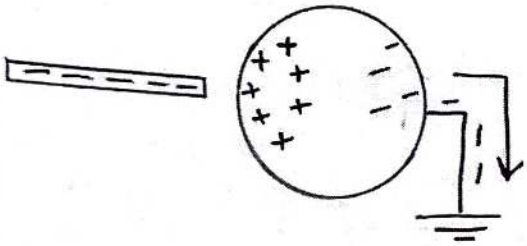
Charging by induction:-



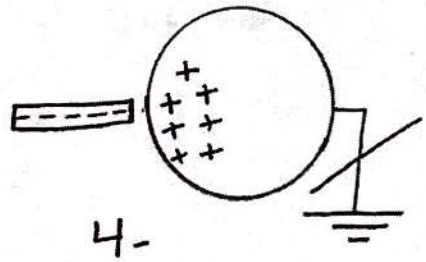
1- neutral conducting sphere.



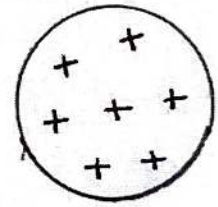
2-



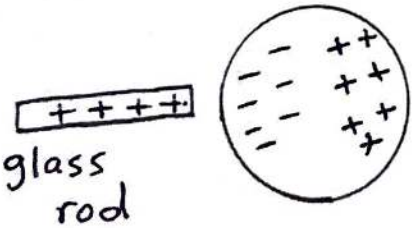
3- ground wire



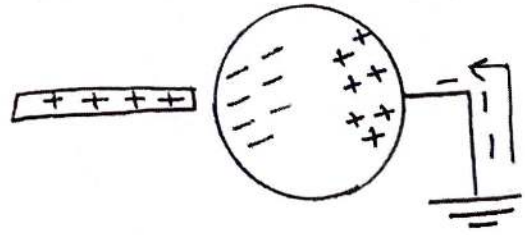
4-



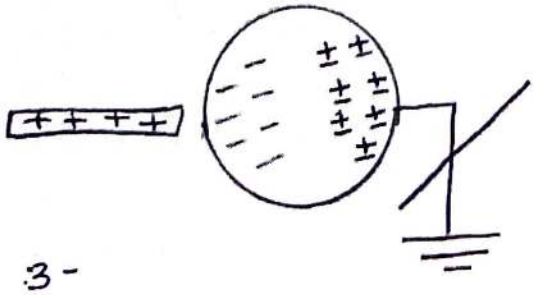
5- Charged conductor



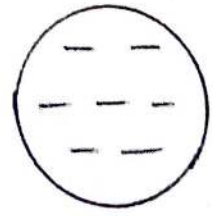
1-



2-



3-



4-

23.3 Coulomb's Law:-

Properties of electric force

$F \propto \frac{1}{r^2}$ $\qquad$ $F \propto q_1 \cdot q_2$

- F is attractive force for opposite sign charges and repulsive force for the same sign charges
- The electric force is conservative force.

$F = \frac{K q_1 q_2}{r^2}$, where

K is Coulomb constant $K = 8.987 \times 10^9 \ N.m^2/C^2$

$K = \frac{1}{4\pi\epsilon_0}$, ϵ_0 is permittivity of vacuum

$\epsilon_0 = 8.85 \times 10^{-12} \ C^2/N \cdot m^2$

Ex(23.1):- Hydrogen atom:-



$r = 5.3 \times 10^{-11} \ m$ $\qquad$ $q_p = 1.6 \times 10^{-19} C$

$q_e = -1.6 \times 10^{-19} C$

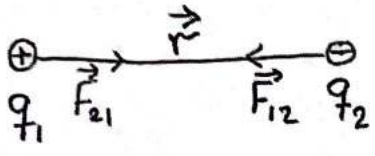
$F_e = \frac{K q_1 q_2}{r^2} = \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{(5.3 \times 10^{-11})^2} = 8.2 \times 10^{-8} \ N$

$$F_g = \frac{G m_e m_p}{r^2}$$

$$= \frac{6.67 \times 10^{-11} \times 9.11 \times 10^{-31} \times 1.67 \times 10^{-27}}{(5.3 \times 10^{-11})^2}$$

$$= 3.6 \times 10^{-47} \text{ N}$$

$$F_g << F_e$$

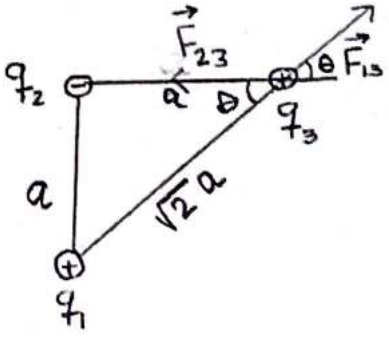


$$\vec{F}_{12} = -\vec{F}_{21}$$

Ex(23.2) Find the resultant Force

$q_1 = q_3 = 5\ \mu C$ $\quad q_2 = -2\ \mu C$ $\quad a = 0.1$ m

Find the resultant force on q_3?



$$F_{13} = \frac{9 \times 10^9 \times (5 \times 10^{-6})^2}{(0.1 \times \sqrt{2})^2} = 11 \text{ N}$$

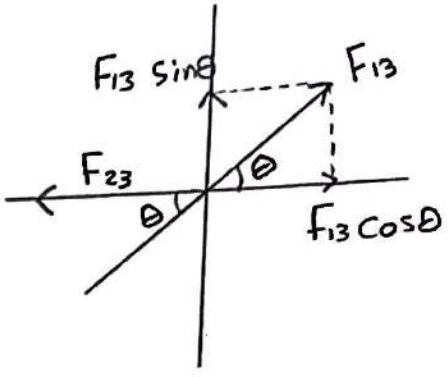
$$\vec{F}_{23} = \frac{9 \times 10^9 \times 5 \times 10^{-6} \times 2 \times 10^{-6}}{(0.1)^2}$$

$$= -9\hat{i} \text{ N}$$

$(\vec{F}_{13})_x = F_{13}\cos\theta$

$= 11 \times \frac{\alpha}{\alpha\sqrt{2}}$

$= \frac{11}{\sqrt{2}}\hat{i}\ N$



$(\vec{F}_{13})_y = F_{13}\sin\theta$

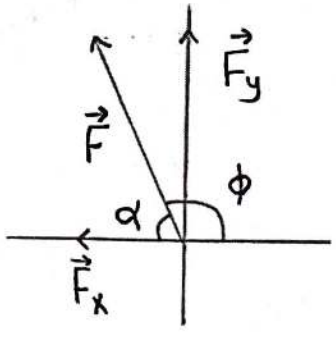
$= 11 \times \frac{\alpha}{\alpha\sqrt{2}} = \frac{11}{\sqrt{2}}\hat{j}\ N$

$\Sigma \vec{F}_x = (\vec{F}_{13})_x + (\vec{F}_{23})$

$= \frac{11}{\sqrt{2}}\hat{i} - 9\hat{i} = -1.1\ N$

$\Sigma F_y = 7.8\hat{j}\ N$

$\vec{F} = F_x\hat{i} + F_y\hat{j} = -1.1\hat{i} + 7.8\hat{j}$



$|\vec{F}| = \sqrt{(F_x)^2 + (F_y)^2}$

$= \sqrt{(-1.1)^2 + (7.8)^2} = 7.9\ N$

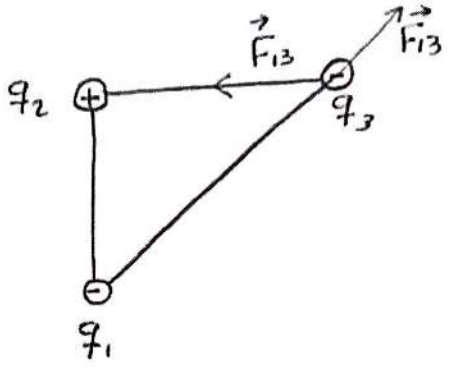
$\tan\alpha = \frac{F_y}{F_x} = \frac{7.8}{-1.1} = -7$

$\alpha = \tan^{-1}(-7) = -82^\circ$

$\phi = 180 - 82 = 98^\circ$

What if?? the signs of all three charges were charged to opposite charge?

the final result for $\Sigma \vec{F}$ would be the same.

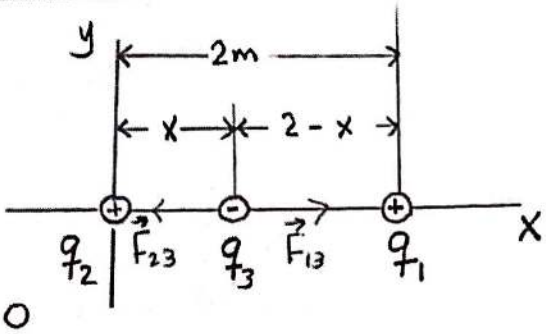


Ex(23.3):- where is the net force zero?

$q_1 = 15\ \mu C$ at $x = 2m$

$q_2 = 6\ \mu C$ at the origion

the resultant force at $q_3 = 0$



what is the x- coordinate of q_3

$$|\vec{F}_{13}| = |\vec{F}_{23}|$$

$$\frac{K q_1 q_3}{r_{13}^2} = \frac{K q_2 q_3}{r_{23}^2}$$

$$\sqrt{\frac{15 \times 10^{-6}}{(2-x)^2}} = \sqrt{\frac{6 \times 10^{-6}}{x^2}}$$

$$\frac{3.87}{2-x} = \frac{2.45}{x}$$

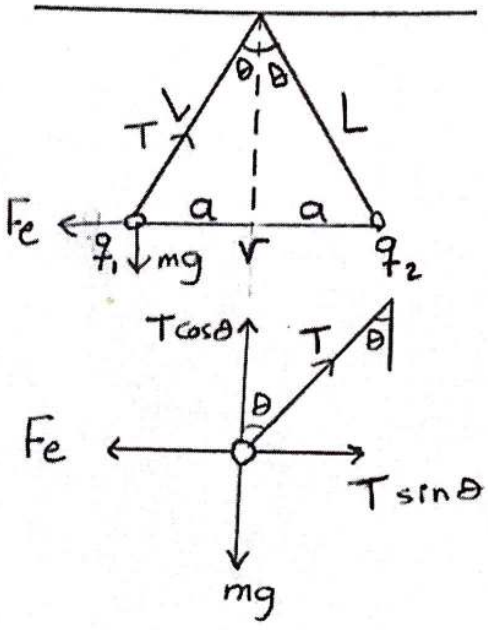
$3.87x = 4.9 - 2.45x$

$x = 0.77\ m$

Ex(23.4):- Find the charges on the spheres?

$m_1 = m_2 = 3\times10^{-2}\ kg \qquad L = 0.15\ m \qquad \theta = 5^\circ$

$q_1 = q_2 = ??$



at equalibrium $\Rightarrow \Sigma F_x = 0$

$\Sigma F_y = 0$

$F_e = T\sin\theta$ ---①

$mg = T\cos\theta$ ---②

$\frac{F_e}{mg} = \tan\theta = \tan 5^\circ$

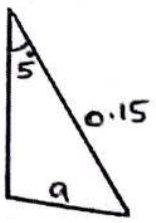
$F_e = 3\times10^{-2}\times10\times0.087$

$= 0.026\ N$

$F_e = \frac{K\,q_1 q_2}{r^2}$

$0.026 = \frac{9\times10^9 \times q^2}{r^2}$

$q^2 = \frac{0.026}{9\times10^9}\,r^2$



$$\sin 5^\circ = \frac{a}{0.15}$$

$$a = 0.013 \text{ m}$$

$$r = 2a = 2 \times 0.013$$

$$= 0.026 \text{ m}$$

$$q^2 = \frac{0.026 \times (0.026)^2}{9 \times 10^9} = 1.97 \times 10^{-15}$$

$$q = 4.42 \times 10^{-8} \text{ C}$$

$$= 44.2 \times 10^{-9} \text{ C}$$

$$= 44.2 \text{ nC}$$

Quick Quiz (23.1), Page 692:-

Quick Quiz 23.1 Three objects are brought close to each other, two at a time. When objects A and B are brought together, they repel. When objects B and C are brought together, they also repel. Which of the following are true? (a) Objects A and C possess charges of the same sign. (b) Objects A and C possess charges of opposite sign. (c) All three objects possess charges of the same sign. (d) One object is neutral. (e) Additional experiments must be performed to determine the signs of the charges.

A repel B , B repel C

answer :- a , C , e

Quick Quiz (23.2), Page 693 :-

Quick Quiz 23.2 Three objects are brought close to one another, two at a time. When objects A and B are brought together, they attract. When objects B and C are brought together, they repel. Which of the following are necessarily true? (a) Objects A and C possess charges of the same sign. (b) Objects A and C possess charges of opposite sign. (c) All three objects possess charges of the same sign. (d) One object is neutral. (e) Additional experiments must be performed to determine information about the charges on the objects.

A attract B, B repel C

suppose that A is +ve then B is -ve

and C is -ve too.

answer:- b , e

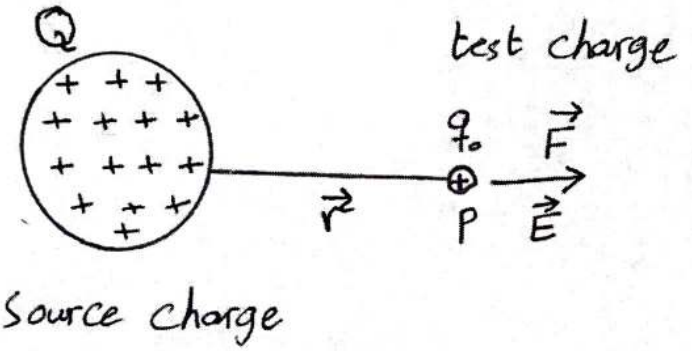
Quick Quiz (23.3), Page 696 :-

Quick Quiz 23.3 Object A has a charge of $+2\ \mu C$, and object B has a charge of $+6\ \mu C$. Which statement is true about the electric forces on the objects? (a) $\vec{F}_{AB} = -3\vec{F}_{BA}$ (b) $\vec{F}_{AB} = -\vec{F}_{BA}$ (c) $3\vec{F}_{AB} = -\vec{F}_{BA}$ (d) $\vec{F}_{AB} = 3\vec{F}_{BA}$ (e) $\vec{F}_{AB} = \vec{F}_{BA}$ (f) $3\vec{F}_{AB} = \vec{F}_{BA}$

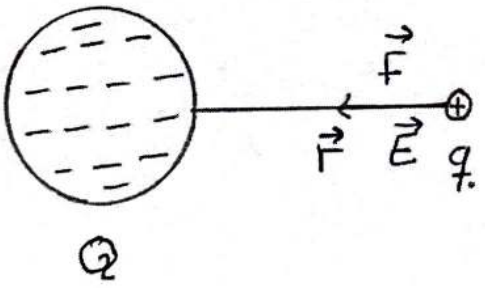
answer :- (b)

23.4 Analysis Model : Particle in a electric field :-

Electric field :- electric force acting on a +ve test charge placed at the point divided by magnitude of the test charge q_0.



Source charge



Q

$$\vec{E} = \frac{\vec{F}}{q_0} \equiv [N/C]$$

$$\vec{F} = \frac{K Q q_0}{r^2} \hat{r}$$

$$\vec{E} = \frac{K Q}{r^2} \hat{r}$$

$\vec{E} = K \sum_i \frac{q_i}{r_i^2} \hat{r}_i$ ≡ total electric field due to a group of charges.

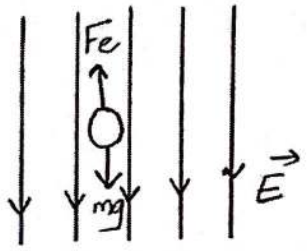
Ex(23.5) A suspended water droplet:-

$m = 3 \times 10^{-12}$ Kg $\vec{E} = -6 \times 10^3 \hat{j}$ N/C

$\Sigma F_y = 0$

$F_e - F_g = 0$

$q(E) - mg = 0$



$$q = \frac{mg}{E}$$

$$= \frac{-3 \times 10^{-12} \times 9.8}{6 \times 10^3}$$

$$= -4.9 \times 10^{-15} \text{ C}$$

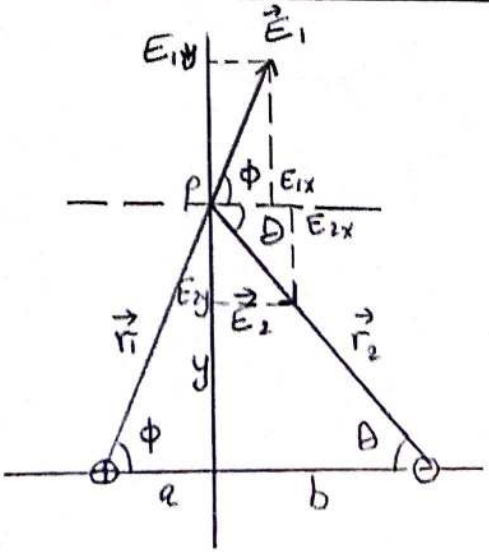
Ex(23.6):- electric field due to two charges:-

A) $E_1 = \frac{K q_1}{r_1^2}$

$r_1 = \sqrt{a^2 + y^2}$

$E_1 = \frac{K q_1}{a^2 + y^2}$

$\vec{E}_1 = \frac{K q_1}{a^2+y^2} \cos\phi \hat{i} + \frac{K q_1}{a^2+y^2} \sin\phi \hat{j}$



$$E_2 = \frac{K q_2}{r_2^2}$$

$$r_2 = \sqrt{b^2 + y^2}$$

$$E_2 = \frac{K q_2}{b^2 + y^2}$$

$$\vec{E}_2 = \frac{K q_2}{b^2+y^2} \cos\theta\, \hat{i} - \frac{K q_2}{b^2+y^2} \sin\theta\, \hat{j}$$

$$E_x = E_{1x} + E_{2x}$$

$$= \frac{K q_1}{a^2+y^2} \cos\phi + \frac{K q_2}{b^2+y^2} \cos\theta$$

$$E_y = E_{1y} + E_{2y}$$

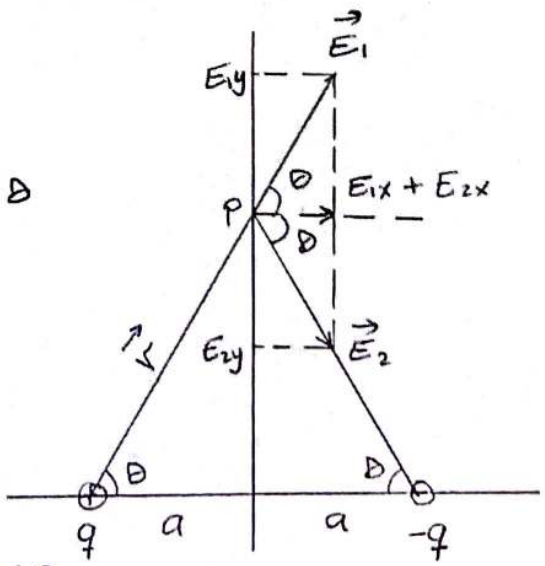
$$= \frac{K q_1}{a^2+y^2} \sin\phi - \frac{K q_2}{b^2+y^2} \sin\theta$$

B) $q_1 = q_2$, $a = b$

$$E_x = \frac{K q}{a^2+y^2} \cos\theta + \frac{K q}{a^2+y^2} \cos\theta$$

$$= \frac{2 K q}{a^2+y^2} \cos\theta$$

$$\cos\theta = \frac{a}{r} = \frac{a}{\sqrt{a^2+y^2}}$$



$$E_x = \frac{2Kq}{a^2+y^2} \cdot \frac{a}{\sqrt{a^2+y^2}}$$

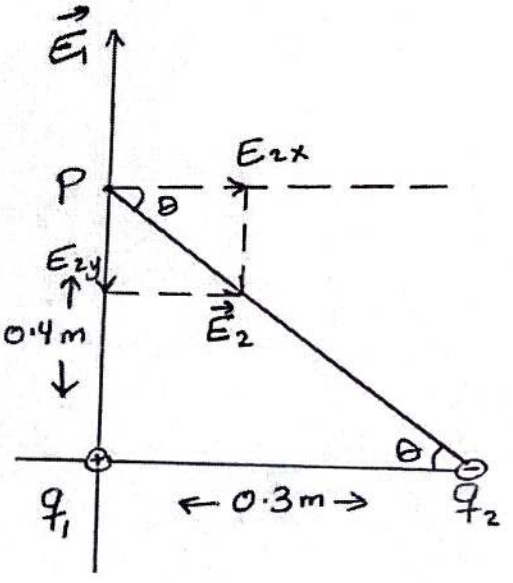
$$E_x = \frac{2Kaq}{(a^2+y^2)^{3/2}}$$

$$E_y = \frac{Kq}{a^2+y^2}\sin\theta - \frac{Kq}{a^2+y^2}\sin\theta$$

$$= 0$$

C) $y \ggg a$

$$E \approx \frac{2Kaq}{y^3}$$



Example:-

$q_1 = 7\mu C \qquad q_2 = -5\mu C$

$$\vec{E}_1 = \frac{Kq_1}{r^2} = \frac{9\times10^{9}\times7\times10^{-6}}{(0.4)^2}$$

$$= 3.94\times10^{5}\,\hat{j}\ N/C$$

$$E_2 = \frac{Kq_2}{r^2} = \frac{9\times10^{9}\times5\times10^{-6}}{(0.5)^2}$$

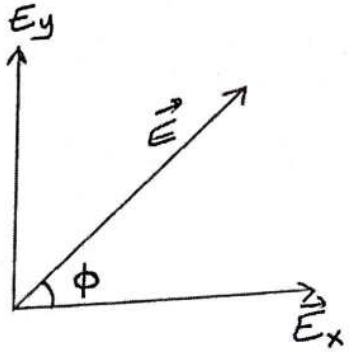
$$= 1.8\times10^{5}\ N/C$$

$$E_{2x} = E_2 \cos\theta = 1.8 \times 10^5 \times \frac{3}{5} = 1.1 \times 10^5 \text{ N/C}$$

$$E_{2y} = E_2 \sin\theta = 1.8 \times 10^5 \times \frac{4}{5} = 1.44 \times 10^5 \text{ N/C}$$

$$\Sigma E_y = (3.94 - 1.44) \times 10^5 = 2.5 \times 10^5 \text{ N/C}$$

$$\vec{E} = [1.1 \times 10^5 \hat{i} + 2.5 \times 10^5 \hat{j}] \text{ N/C}$$



$$|\vec{E}| = \sqrt{(1.1 \times 10^5)^2 + (2.5 \times 10^5)^2}$$

$$= 2.7 \times 10^5 \text{ N/C}$$

$$\Phi = \tan^{-1}\left(\frac{E_y}{E_x}\right)$$

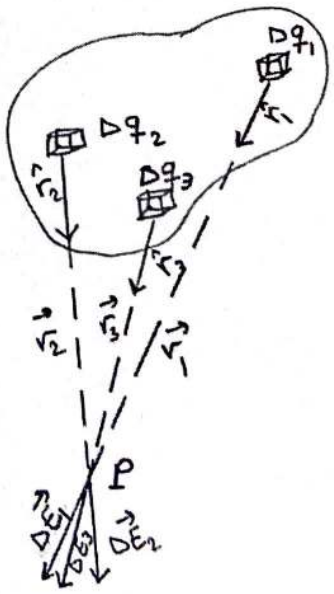
$$= \tan^{-1}\left(\frac{2.5}{1.1}\right) = 66^\circ$$

23.5 Electric field of a continuous charge distribution:-

$$\Delta\vec{E} = \frac{K\,\Delta q}{r^2}\,\hat{r}$$

r : distance from the element charge to the point P.

$\hat{r}$:- unit vector directed from element charge toward P.



total electric field

$$E = K \lim_{\Delta q_i \to 0} \sum_i \frac{\Delta q_i}{r_i^2}\,\hat{r}_i \Rightarrow$$

$$\vec{E} = K_e \int \frac{dq}{r^2} \hat{r}$$

if a charge is uniformly distributed through a volume V, ρ is the volume charge density

$$\rho = \frac{Q}{V} \equiv [C/m^3]$$

Or a surface of area A, σ is the surface charge density

$$\sigma = \frac{Q}{A} \equiv [C/m^2]$$

or along a line of length L, λ is the length charge density

$$\lambda = \frac{Q}{L} \equiv [C/m]$$

but when a charge is nonuniformly distributed

over a volume $dQ = \rho \, dV$

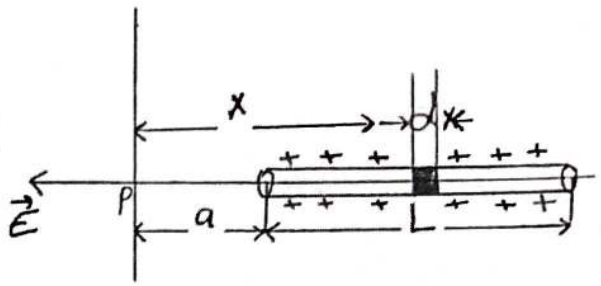
a surface $dQ = \sigma \, dA$

a length $dQ = \lambda \, dL$

Ex(23.7):- The electric field due to a charged rod:

Q :- total charge

λ :- Linear charge density



$$dq = \lambda\, dx$$

$$dE = \frac{K\, dq}{x^2} = \frac{K \lambda\, dx}{x^2}$$

$$E = K \int_{a}^{L+a} \frac{\lambda\, dx}{x^2}$$

$$= K\lambda \int_{a}^{L+a} x^{-2}\, dx = K\lambda \left[\frac{-1}{x}\right]_{a}^{L+a}$$

$$= K\lambda \left[\frac{1}{a} - \frac{1}{a+L}\right]$$

$$= K\lambda \left[\frac{a+L-a}{a(a+L)}\right]$$

$$= K\lambda \left[\frac{L}{a(a+L)}\right]$$

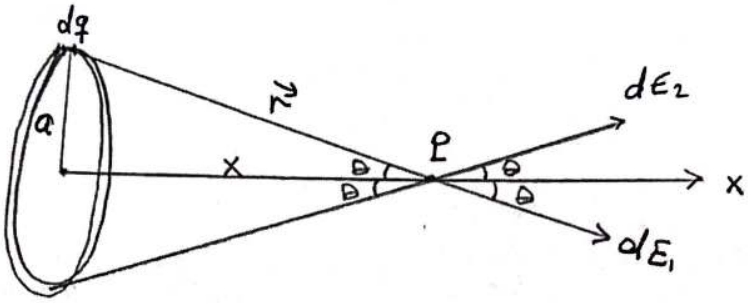
$$= K \frac{Q}{L} \left[\frac{L}{a(a+L)}\right]$$

$$E = \frac{K Q}{a(a+L)}$$

when $a >>> L \Rightarrow E = \frac{KQ}{a^2}$

the rod becomes point charge

Ex (23.8):- the electric field of a uniform ring of charge:-



$$dE = \frac{K\,dq}{r^2} = \frac{K\,dq}{x^2+a^2}$$

$dE_y = 0$, by symmetry

$$dE_x = dE\cos\theta$$

$$= \frac{K\,dq}{x^2+a^2}\cdot\frac{x}{(x^2+a^2)^{1/2}}$$

$$= \frac{K\,x\,dq}{(x^2+a^2)^{3/2}}$$

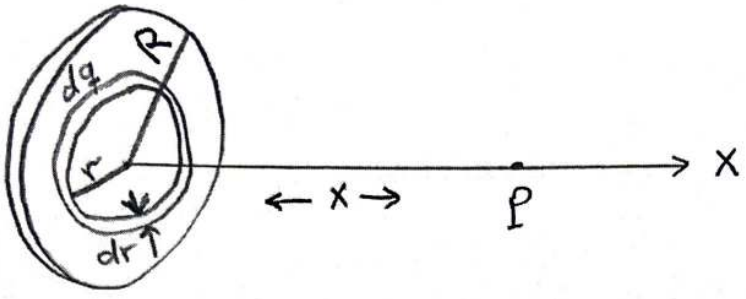
$$E = \int dE_x = K\int \frac{x\,dq}{(x^2+a^2)^{3/2}}$$

$$= \frac{K\times x\times Q}{(x^2+a^2)^{3/2}}$$

for $x >>> a \Rightarrow E = \frac{KQ}{x^2}$ (ring becomes point charge)

for $x = 0 \Rightarrow E = 0$

Ex(23.9):- The electric field of auniformly charged disk:-



$$dq = \sigma\, dA = \sigma (2\pi r\, dr)$$

$$dE_x = \frac{K\, x\, dq}{(x^2+r^2)^{3/2}}$$, from previous example r instead of a

$$E_x = \int_0^R \frac{K\, x\, (2\pi r)\sigma\, dr}{(x^2+r^2)^{3/2}} = 2\pi\sigma K x \int_0^R \frac{r\, dr}{(x^2+r^2)^{3/2}}$$

let $u = x^2+r^2$

$$du = 2r\, dr \Rightarrow dr = \frac{du}{2r}$$

$$\int \frac{r\, dr}{(x^2+r^2)^{3/2}} = \int \frac{r}{u^{3/2}} \cdot \frac{du}{2r} = \frac{1}{2}\int u^{-3/2}\, du$$

$$= \frac{1}{2}(-2u^{-1/2}) = \frac{-1}{u^{1/2}} = \frac{-1}{(x^2+r^2)^{1/2}}\Big]_0^R$$

$$= -\left[\frac{1}{(x^2+R^2)^{1/2}} - \frac{1}{x}\right]$$

$$= \frac{1}{x} - \frac{}{(x^2+R^2)^{1/2}}$$

$$E_x = 2\pi\sigma K x\left[\frac{1}{x} - \frac{1}{(x^2+R^2)^{1/2}}\right]$$

$$= 2\pi\sigma K\left[1 - \frac{x}{(x^2+R^2)^{1/2}}\right]$$

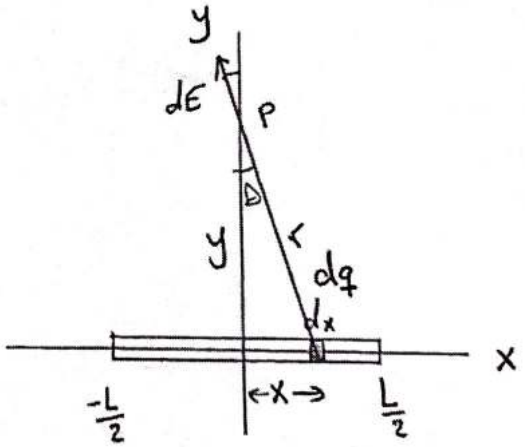
when P too near the surface ; $R \gg x$

$$E = 2\pi\sigma K$$

$$= \frac{2\pi\sigma}{4\pi\epsilon_0} = \frac{\sigma}{2\epsilon_0}$$

for infinite sheet $\Rightarrow$ uniform electric field

Example:-



$$dE = \frac{K\,dq}{r^2}$$

$$= \frac{K\lambda\,dx}{x^2+y^2}$$

$$dE_y = dE\cos\theta$$

$$= \frac{K\lambda\,dx}{x^2+y^2} \cdot \frac{y}{\sqrt{x^2+y^2}}$$

$$= \frac{K\lambda y\,dx}{(x^2+y^2)^{3/2}}$$

$$E_y = \int_{-\frac{L}{2}}^{\frac{L}{2}} K\lambda y\,\frac{dx}{(x^2+y^2)^{3/2}}$$

$$E_y = \frac{K\lambda y\, x}{y^2\sqrt{x^2+y^2}}\Bigg]_{-\frac{L}{2}}^{L/2}$$

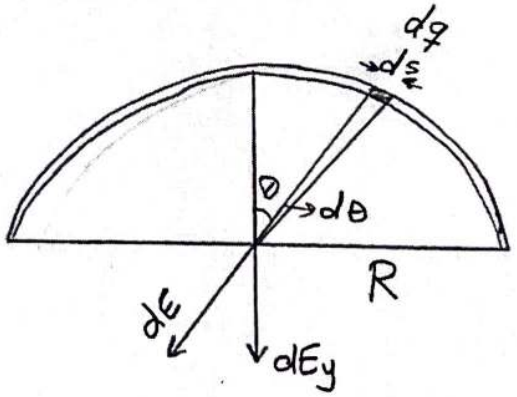
$$= \frac{K\lambda L}{y\sqrt{(\frac{L}{2})^2+y^2}}$$

$$E_y = \frac{K Q}{y\sqrt{(\frac{L}{2})^2+y^2}}$$

for $y >>> L \Rightarrow E_y = \frac{KQ}{y^2}$

for $y <<< L \Rightarrow E_y = \frac{K\lambda L}{y\,\frac{L}{2}}$

$$E_y = \frac{2K\lambda}{y}$$, for infinite rod.



Example:-

$$dE = \frac{K\,dq}{R^2}$$

$$dE_y = dE\cos\theta$$

$$= \frac{K\,dq}{R^2}\cos\theta$$

$$\sin(d\theta) = \frac{ds}{R} = d\theta \quad \text{(for small angle)}$$

$$\therefore ds = R\,d\theta$$

$$dq = \lambda\, ds$$
$$= \lambda R\, d\theta$$

$$dE_y = \frac{K\, dq}{R^2}\cos\theta$$
$$= \frac{K\lambda}{R}\cos\theta\, d\theta$$

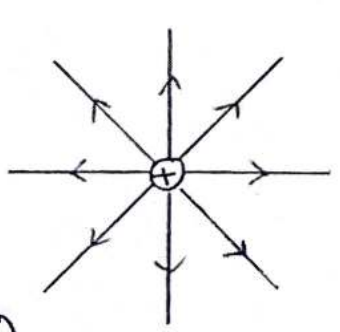
$$E_y = \frac{2K\lambda}{R}\int_0^{\pi/2}\cos\theta\, d\theta = \frac{2K\lambda}{R}\sin\theta\Big]_0^{\pi/2}$$

$$= \frac{2K\lambda}{R}(1-0) = \frac{2K\lambda}{R}$$

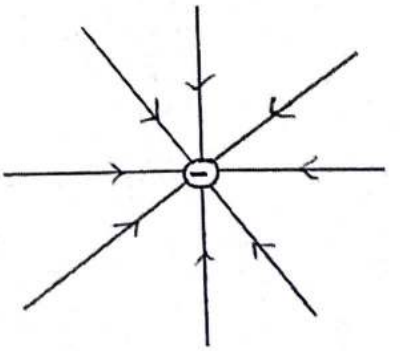
in general for arc $E = \frac{2K\lambda}{R}\sin\theta$

where $\theta = \frac{1}{2}$ angle of arc.

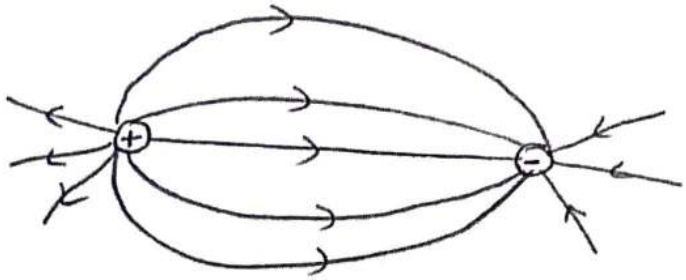
[23.6] Electric Field Lines:-



① For +ve charge, $\vec{E}$ lines are directed radially outward.



For -ve charge, $\vec{E}$ lines are directed radially inward.

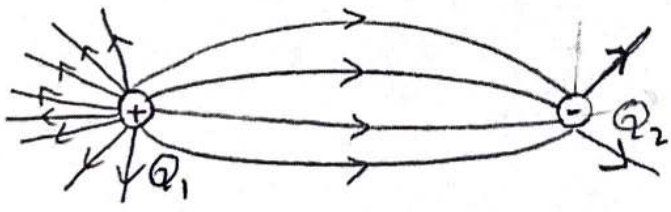


② the direction of $\vec{E}$ line is the direction of the tangent for curved line and the direction of the straight line.

③ No two field Lines can cross.

④ the number of electric field Lines are directly Proportional to the magnitude of the charge.

Ex for $Q_1 = 2q$ and $Q_2 = q$ draw the $\vec{E}$ lines diagram.



$N_1 = 12$ $N_2 = 6$

$$\frac{N_1}{N_2} = \frac{Q_1}{Q_2} = \frac{2q}{q}$$

$$N_1 = 2N_2$$

23.7 Motion of a charged particle in auniform $\vec{E}$

for particle of charge q and mass m, $\vec{E}$ act on the charge by force qE

$$\Sigma \vec{F} = m\vec{a} = q\vec{E}$$

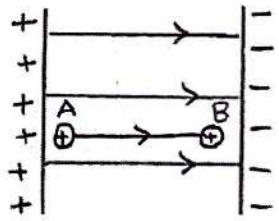
$$\vec{a} = \frac{q\vec{E}}{m}$$

$+q$ accelerates in the direction of $\vec{E}$.

$-q$ accelerates in the opposite direction of $\vec{E}$.

EX(23.10):- an accelerating positive charge:-

$V_A = 0 \qquad X_A = 0$



a) $\vec{a} = \frac{q\vec{E}}{m} = \text{constant}$

$$\Delta X = \cancel{V_A t}^{\,0} + \frac{1}{2}at^2 = X_B - \cancel{X_A}^{\,0}$$

$$X_B = \frac{1}{2}at^2 = \frac{1}{2}\frac{qE}{m}t^2$$

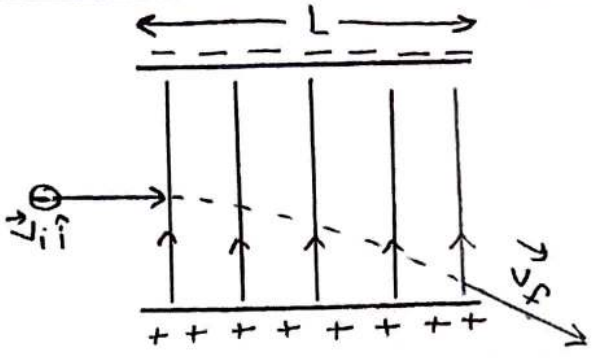
$$V_B^2 = \cancel{V_A^2}^{\,0} + 2a(X_B - X_A) = 0 + 2a(d-0) = 2ad$$

$$V_B = \sqrt{2ad} = \sqrt{\frac{2qEd}{m}}$$

b) $W = \Delta K \Rightarrow F\Delta X = K_B - K_A = \frac{1}{2}mV_B^2 - 0$

$$V_B = \sqrt{\frac{2F\Delta X}{m}} = \sqrt{\frac{2qEd}{m}}$$

Ex(23.11):- An accelerated electron:-



$v_i = 3 \times 10^6$ m/s

$E = 200$ N/C

$L = 0.1$ m

at $t = 0$

$\vec{v} = v_i \hat{i}$

$a_x = 0$; $F_x = 0$

$\vec{a}_y = -\frac{eE}{m_e} \hat{j}$

$v_{xi} = v_i$, $v_{yi} = 0$

$=$ constant

$v_y = a_y t = -\frac{eE}{m_e} t$

$x_f = v_i t \Rightarrow t = \frac{x_f}{v_i}$

$y_f = \frac{1}{2} a_y t^2 = -\frac{1}{2} \frac{eE}{m_e} t^2 = -\frac{1}{2} \frac{eE}{m_e} \left(\frac{x_f}{v_i}\right)^2$

a) $a_y = -\frac{eE}{m_e} \hat{j} = -\frac{(1.6 \times 10^{-19}) \times 200}{9.11 \times 10^{-31}}$

$= -3.51 \times 10^{13} \hat{j}$ m/s^2

b) $x_f = x_i + v_x t + \frac{1}{2} a_x t^2$

$x_f - x_i = v_x t \Rightarrow t = \frac{\Delta x}{v_x} = \frac{0.1}{3 \times 10^6} = 3.33 \times 10^{-8}$ sec

c) $y_f = \frac{1}{2} a_y t^2 = -\frac{1}{2} (3.51 \times 10^{13})(3.33 \times 10^{-8})^2$

$= -0.0195$ m

$= -1.95$ cm